

Efficient Fischer-Tropsch to light olefins over iron-based catalyst with low methane selectivity and high olefin/paraffin ratio

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ABSTRACT

Biomass, biogas, and other carbon-based matters can be processed to generate syngas, which may be catalytically converted to useful hydrocarbons through Fischer-Tropsch reactions. A series of Fe/K catalysts were synthesized using high temperature treated supports with different ratios of zinc aluminate spinel to alumina and were examined by XRD, ICP, FESEM, CO₂-TPD, nitrogen physisorption, and syngas conversion tests. At a total pressure of 2 MPa ($\approx$ 20 bar) and a CO conversion of 38.3%, the C₂, C₃ and C₄ olefin/paraffin ratios of catalyst 15%Fe/2% K₂O/83%(ZnAl₂O₄·1Al₂O₃) exhibit impressive values of 9.8, 17.7 and 13.0, respectively. Meanwhile, its C₂-C₄ HC selectivity reaches a considerable value of 52.1%, where its C₂, C₃ and C₄ olefin fractions attain ultra-high values of 90.7%, 94.7%, and 92.9%, respectively. Its methane HC selectivity is also successfully restrained to be lower than 9% even at the highest CO conversion.

1. Introduction

Biomass, biogas, and other carbon-based matters such as coal, carbon dioxide as well as natural gas, can be processed to generate syngas, a mixture of carbon monoxide and hydrogen, through dry reforming, steam reforming, and water gas shift [1–4]. Syngas, also called synthetic gas, may be catalytically converted to oil, wax, light olefins, or oxygenates [5–24].

Light olefins include ethylene (C₂), propylene (C₃), butylene (C₄), and is usually presented as C₂-C₄ [17]. As extensively used bulk chemicals for the production of plastics, detergents, drugs, cosmetics and solvents, light olefins may be prepared by petroleum processing or direct/indirect conversion of syngas.

Functional inorganic solid catalysts [10,13–24] may be utilized for conversion of syngas via a process of Fischer-Tropsch to light olefins (FTO). At 20 bar and 340 °C, the C₂-C₄ hydrocarbon (HC) selectivity of famous S/Na promoted Fe-based FTO catalyst of de Jong et al. reaches a maximum value of 51% with a C₂-C₄ O/P ratio of 2.7, however, their methane HC selectivity is as high as 19% [14]. As an unfavorable product for FTO process, methane (CH₄) is a completely hydrogenated chemical, the HC selectivity of which ought to be intensely affected by catalyst's ability of hydrogenation. In this article, we report our FTO catalyst that is able to effectively restrain the methane HC selectivity by

suppressing of hydrogenation.

2. Experimental section

2.1. Preparation of materials

The (Al)-O-(Zn) series of support materials are prepared by precipitation/coprecipitation using sodium bicarbonate and the nitrates of elements, of which the dosages follow the ratios of elemental compositions in Table 1. Syntheses of support materials (SP0, SP1, SP5, SP ∞) are conducted in a sequence of precipitation/coprecipitation, extrusion, drying, and subsequent high temperature calcination at 1200 °C for 24 h. For instance, support SP1 is coprecipitated with a precipitant (i.e., NaHCO₃) and a solution of zinc/aluminum nitrates at 40 °C under vigorous stirring, filtered after an aging under stirring for 1 h, thoroughly washed with water, and then partially dried at 80 °C. The half-dried filter cake was mixed with sesbania gum powder (i.e., an organic agent in an amount of 5% by weight for assisting extrusion and will be burnt off during subsequent calcination process) and extruded, air-dried at RT, desiccated at 110 °C for 2 h, and then calcined in an air-atmosphere tubular furnace at 1200 °C for 24 h. The received support material is denoted as SP1. Other supports are prepared similarly. Unreduced catalysts N1, N5, REF0 and REF ∞ are respectively

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Table 1
Nomenclature and crystalline phases of samples.*

Sample code	nominal composition (metal/support) (wt)	x value #	crystalline phases in XRD patterns of supports or freshly reduced catalysts	crystalline phases in XRD patterns of used catalysts
N1	15%Fe/2%K ₂ O/83% (ZnAl ₂ O ₄ ·1Al ₂ O ₃)	1	Fe ⁰ (minor), FeO (major), ZnAl ₂ O ₄ (major), θ-Al ₂ O ₃ (minor)	Fe ₅ C ₂ (minor), ZnAl ₂ O ₄ (major), θ-Al ₂ O ₃ (minor)
N5	15%Fe/2%K ₂ O/83% (ZnAl ₂ O ₄ ·5Al ₂ O ₃)	5	Fe ⁰ (major), FeO (minor), α-Al ₂ O ₃ (major), β-Al ₂ O ₃ (minor), ZnAl ₂ O ₄ (minor)	Fe ₅ C ₂ (minor), α-Al ₂ O ₃ (major), β-Al ₂ O ₃ (minor), ZnAl ₂ O ₄ (minor)
REF _∞	15%Fe/2%K ₂ O/83%Al ₂ O ₃	∞	Fe ⁰ (major), β-Al ₂ O ₃ (minor), α-Al ₂ O ₃ (few)	β-Al ₂ O ₃ (major), α-Al ₂ O ₃ (minor)
REF0	15%Fe/2%K ₂ O/83%ZnAl ₂ O ₄	0	Fe ⁰ (major), ZnAl ₂ O ₄ (major)	Fe ₅ C ₂ (minor), ZnAl ₂ O ₄ (major)
SP1	100% (ZnAl ₂ O ₄ ·1Al ₂ O ₃)	1	ZnAl ₂ O ₄ (major), θ-Al ₂ O ₃ (minor), β-Al ₂ O ₃ (few)	N/A
SP5	100% (ZnAl ₂ O ₄ ·5Al ₂ O ₃)	5	α-Al ₂ O ₃ (major), β-Al ₂ O ₃ (minor), ZnAl ₂ O ₄ (minor)	N/A
SP _∞	100%Al ₂ O ₃	∞	β-Al ₂ O ₃ (major), α-Al ₂ O ₃ (minor)	N/A
SP0	100%ZnAl ₂ O ₄	0	ZnAl ₂ O ₄	N/A

* N1, N5, REF_∞ and REF0 are “catalyst” samples. SUP1, SUP5, SUP_∞ and SUP0 are “support” samples. # $x = n(\text{Al}_2\text{O}_3)/n(\text{ZnAl}_2\text{O}_4)$, i.e., molar ratio of nominal Al₂O₃ to nominal ZnAl₂O₄.

synthesized by incipient wetness of supports SP1, SP5, SP0 and SP_∞ with aqueous solutions of K₂CO₃, drying, another incipient wetness with aqueous solutions of ammonium iron citrate, another drying and subsequent calcination at 350 °C for 4 h in air. Catalysts REF0 and REF_∞ are utilized for reference purpose. In this article, SP0 and SP_∞ are reference supports, whilst REF0 and REF_∞ are reference catalysts. Generally, K₂O % is employed to simplify the expression of potassium content in the catalyst composition of potassium carbonate application [12,15]. High temperature annealing at 1200 °C is usually employed in the literature [13,14] to transform alumina phases to α-Al₂O₃ phase, through which the surface acidic sites may be significantly decreased via the removal of surface hydroxyls and, meanwhile, the surface basic sites may be modulated to alter the selectivities of products. Therefore, this article utilizes high temperature annealing at 1200 °C to prepare solid-base type catalysts. More details about the preparation of each material are described in Supporting Information.

2.2. Characterization

XRD, ICP, nitrogen physisorption, FESEM, and CO₂-TPD were conducted for materials characterization. More details about the characterization are described in Supporting Information.

2.3. Catalytic performance tests

Catalytic CO hydrogenation reactions are carried out with a fixed-bed tubular reactor. Catalysts (i.e., in oxide status, 40–60 mesh) are loaded and then *in-situ* reduced at 550 °C for 6 h by 50%H₂/50%Ar at 0.1 MPa. A pre-mixed syngas (H₂/CO/Ar = 45/45/10, by mole%) is applied over the *in-situ* reduced catalysts for CO hydrogenation reaction at a total pressure of 2.0 MPa as well as a space velocity (SV) of 1500 mL·g⁻¹·h⁻¹. The GC analysis method is similar to our previous reports [15,17,25–28]. More details about the catalytic tests are described in Supporting Information.

3. Results and discussion

3.1. Structure features

The structure features of supports, freshly reduced catalysts, as well as used catalysts after catalytic reaction tests are analyzed by power XRD, of which the results are presented in Table 1, Figs. 1 and 2 and Supporting Information Figs. S1–S5. As listed in Table 1, a value of x represents the molar ratio of nominal Al₂O₃ to nominal ZnAl₂O₄ in the nominal composition of each material. The x value of support materials SP0, SP1, SP5 and SP_∞ equals to 0, 1, 5 and ∞, respectively. The freshly reduced catalysts for instrumental analyses are prepared by loading calcined oxide-status catalysts in alundum boats in a horizontal tubular furnace and reducing at 550 °C for 6 h under a blanket of a reductive gas mixture of 50%H₂/50%Ar.

As listed in Table 1, the nominal composition of reference support material SP_∞ is 100% Al₂O₃. As shown in Fig. 1b and Fig. S1, the major crystalline phase of support SP_∞ is β-Al₂O₃ (JCPDS 31-1263, 10-414, 51-769), while its minor crystalline phase is α-Al₂O₃ (JCPDS 05-0712, 50-0741). Both β-Al₂O₃ and α-Al₂O₃ are refractory phases. Beta-alumina includes β-Al₂O₃ and β'-Al₂O₃. Beta-alumina belongs to hexaaluminates (i.e., hexagonal aluminates) which exhibit a layer structure composed of conduction layers and spinel blocks that are stacked alternately. Na⁺, Ba²⁺, K⁺ etc. are positively charged ions and can be introduced to the conduction layer of beta-alumina via high temperature calcinations [29]. Measured by ICP-OES, the sodium content of support SP_∞ is 2.49% (Table S1). The sodium element in support SP_∞ comes from the residue of the precipitant (i.e., NaHCO₃). Unlike potassium element, sodium element alone has only feeble promoting effect, although sodium plus sulfur (i.e., a combination of Na/S) may generate a synergetic promoting effect over iron catalysts [13,14]. Above facts indicate that a part of Al–O species in uncalcined precursor of support SP_∞ can react with sodium through solid-state reaction at a high temperature of 1200 °C to form refractory Na-β-Al₂O₃ phase (i.e., typically NaAl₁₁O₁₇). Meanwhile, the residual, unreacted Al–O species in uncalcined precursor of support SP_∞ may build an α-Al₂O₃ phase. For a certain XRD pattern of a sample, the relatively strongest peaks are described as “major-level”, the relatively medium peaks are described as “minor-level”, and the relatively weakest peaks are described as “few-level”. As shown in Fig. 1 and Table 1, freshly reduced reference catalyst REF_∞ comprises Fe⁰ (a major-level phase, JCPDS 06-0696), β-Al₂O₃ (a minor-level phase) and α-Al₂O₃ (a few-level phase). As shown in Fig. 2, Fig. S5 and Table 1, after catalytic tests, used reference catalyst REF_∞ consists of β-Al₂O₃ (a major-level phase) and α-Al₂O₃ (a minor-level phase), while its status of iron species becomes X-ray amorphous.

The crystalline phase of reference support material SP0 is simply ZnAl₂O₄ (JCPDS 05-0669) phase (Fig. 1&S1). Determined by ICP-OES, the sodium content of support SP0 is 1.25% and is less than that of support SP_∞ (Table S1). Freshly reduced reference catalyst REF0 contains Fe⁰ and ZnAl₂O₄ phases, both exhibiting strong diffraction peaks (Fig. 1). After catalytic tests, used reference catalyst REF0 is composed of Fe₅C₂ (a minor-level phase, JCPDS 51-0997) and ZnAl₂O₄ (a major-level phase) (Table 1, Fig. 2, Fig. S5). The XRD patterns of supports SP_∞ and SP0 [17] are permitted by copyright of John Wiley & Sons.

Support material SP1 is composed of ZnAl₂O₄ (a major-level phase), θ-Al₂O₃ (a minor-level phase) and β-Al₂O₃ (a few-level phase) (Fig. S1, Table 1). The sodium content of support SP1 is 1.19% and is less than that of support SP_∞ (Table S1). Freshly reduced catalyst N1 comprises Fe⁰ (a minor-level phase), FeO (a major-level phase), ZnAl₂O₄ (a major-level phase) and θ-Al₂O₃ (a minor-level phase) (Fig. S2). Although crystalline β-Al₂O₃ phase is observable in support SP1, it becomes unobservable in the XRD pattern of freshly reduced catalyst N1. After catalytic tests, used catalyst N1 is composed of Fe₅C₂ (a minor-level phase), ZnAl₂O₄ (a major-level phase) and θ-Al₂O₃ (a minor-level phase) (Fig. S2).

Support material SP5 comprises α-Al₂O₃ (a major-level phase),

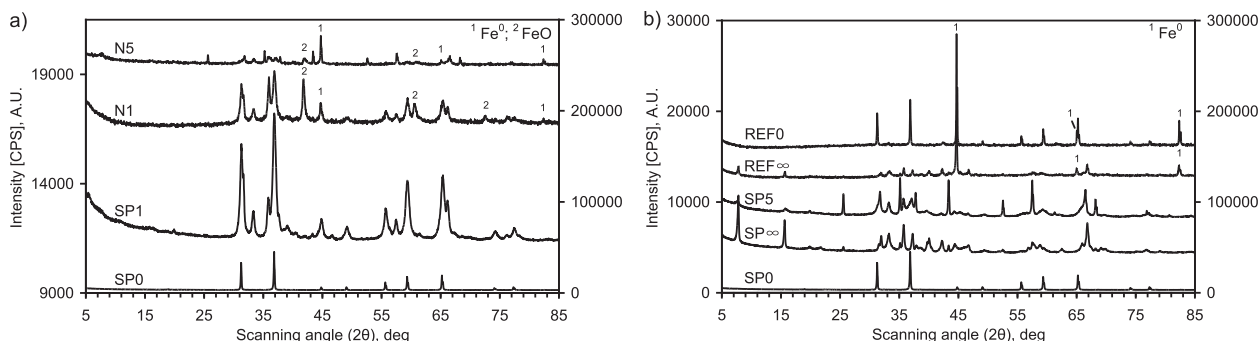


Fig. 1. X-ray diffraction (XRD) patterns of supports and freshly reduced catalysts. Pattern “SP0” follows the scale of the right ordinate axis, while others follow the scale of the left ordinate axis.

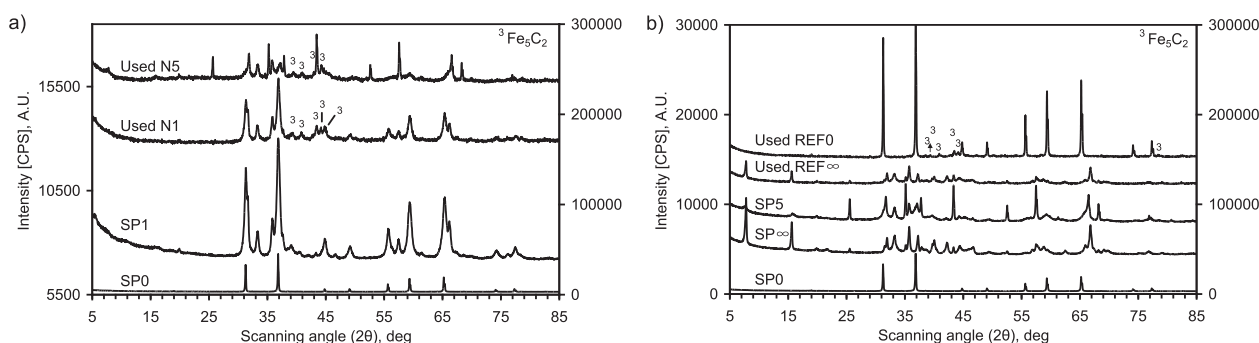


Fig. 2. X-ray diffraction (XRD) patterns of supports and used catalysts discharged after CO hydrogenation tests. Pattern “SP0” follows the scale of the right ordinate axis, while others follow the scale of the left ordinate axis.

β - Al_2O_3 (a minor-level phase) and ZnAl_2O_4 (a minor-level phase) (Fig. S3, Table 1). The sodium content of support SP5 is 1.21% and is less than that in support SP ∞ (Table S1). Freshly reduced catalyst N5 contains Fe^0 (a major-level phase), FeO (a minor-level phase), α - Al_2O_3 (a major-level phase), β - Al_2O_3 (a minor-level phase) and ZnAl_2O_4 (a minor-level phase) (Fig. S4). After catalytic tests, used catalyst N5 is composed of Fe_5C_2 (a minor-level phase), α - Al_2O_3 (a major-level phase), β - Al_2O_3 (a minor-level phase) and ZnAl_2O_4 (a minor-level phase) (Fig. S4).

The location of sodium element of alumina-containing catalysts N1, N5 and REF ∞ is most likely in the 1200 °C annealed alumina phases (β - Al_2O_3 and θ - Al_2O_3) which incorporate sodium atoms as part of themselves, while the sodium of alumina-free catalyst REF0 is X-ray amorphous. The location of zinc element of catalysts REF0, N1 and N5 is in the ZnAl_2O_4 phase, while catalyst REF ∞ is Zn-free.

3.2. Texture features

The mesoporosity texture of freshly reduced catalysts N1, N5, REF0 and REF ∞ is determined through nitrogen sorption tests. As shown in Fig. 3a, all catalysts should be ascribed to mesoporous adsorbents because all of their isotherm plots are type IV according to the assortment of IUPAC [30]. The isotherm plots also disclose that all mesoporosities of these four catalysts are formed by clusters of plate-like particles which generate slit-shape pores, because their hysteresis loops can be assigned to type H3 according to the classification of IUPAC [30]. The pore size distribution (Fig. 3b) of catalyst N5 is different from other catalysts in the pore size range smaller than 3 nm. The BET specific surface areas of catalysts REF0, N1, N5 and REF ∞ are respectively 1.1, 10.7, 19.6 and 4.7 m^2/g , indicating that a catalyst with a binary phase support (i.e., N1 & N5 in which the support contains both zinc aluminate and alumina) may have larger specific surface area than a catalyst with a single phase support (i.e., REF0 & REF ∞ in which the support contains

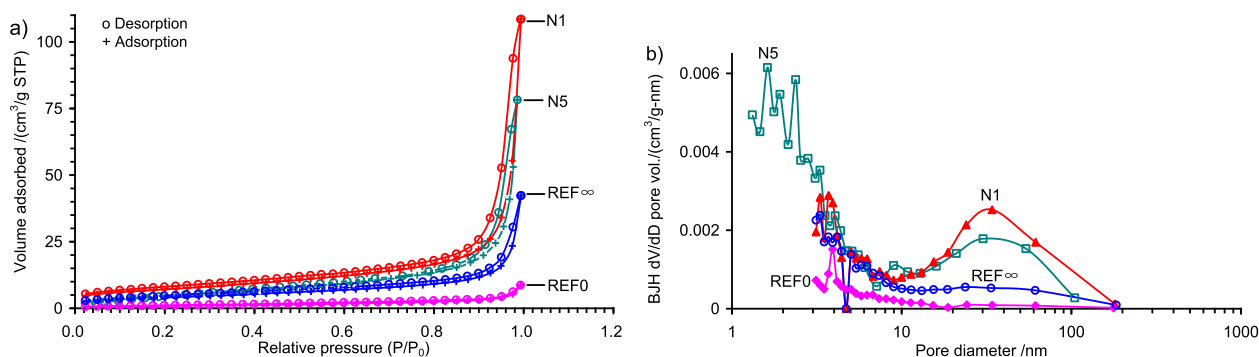


Fig. 3. Results of N_2 sorption measurement of freshly reduced catalysts. (a) Isotherm plot. (b) BJH pore size distribution derived from the desorption branch of isotherm plot.

either zinc aluminate only or alumina only). The major pore sizes of catalysts REF0, N1, N5 and REF ∞ are respectively 3.9, 3.7, 1.6 and 3.3 nm. The total pore volume values of catalysts REF0, N1, N5 and REF ∞ are respectively 0.013, 0.168, 0.121 and 0.066 cm³/g. The average pore diameters (4 V/A) of catalysts REF0, N1, N5 and REF ∞ are 48.9, 62.9, 24.7 and 56.0 nm, respectively.

3.3. Electron microscopy studies

Field-emission scanning electron microscopy (FESEM) is a powerful tool for morphology observation of solid materials. Catalyst REF ∞ is Zn-free and contains mostly (83%) alumina, some (15%) iron plus a little bit (2%) potassium species. Basing on Fig. 4f and Table 1, catalyst REF ∞ mainly consists of partially transparent flakes which can be assigned to β -Al₂O₃ phase. The small quantity of cubic particles of catalyst REF ∞ is ascribed to α -Al₂O₃ phase. The tiny irregular-shape particles (i.e., ca. 30 nm) deposited on alumina phases of catalyst REF ∞ may be Fe⁰ and potassium species. The α -Al₂O₃ cubic particles have an edge length of ca. 100–300 nm. The β -Al₂O₃ flakes of catalyst REF ∞ look irregular, having a thickness of ca. 80 nm and an edge length of ca. 200–1500 nm.

The support of catalyst REF0 is a pure phase material. Catalyst REF0 comprises mostly (83%) zinc aluminate, some (15%) iron plus a little bit (2%) potassium species. According to Fig. 4c and Table 1, the great mass of catalyst REF0 is layered stacks, while some cubes or cuboids also exist. The layered stacks and cubes/cuboids may be ascribed to zinc aluminate phase, in which the cubes/cuboids are caused by Ostwald Ripening. The particles (i.e., ca. 80 nm) deposited on zinc aluminate phase of catalyst REF ∞ may be Fe⁰ and potassium species.

Basing on Fig. 4a&4b, catalyst N1 consists of aggregates of tiny particles, some cubes/cuboids, and a little irregular-shape plates. The tiny particles of catalyst N1 exhibits a size range of 30–120 nm and are perhaps θ -Al₂O₃, unripened zinc aluminate as well as iron species. The cube/cuboid-shape particles of catalyst N1 have an edge size of 400–800 nm and are thoroughly ripened zinc aluminate phase. The

irregular-shape plates have an edge size of 300–800 nm and are possibly half ripened zinc aluminate phase.

According to Fig. 4d&4e, catalyst N5 comprises partially transparent flakes, nontransparent cubes, and tiny particles. The partially transparent flakes of catalyst N5 have an edge size of 200–1000 nm and are possibly β -Al₂O₃ phase and half ripened ZnAl₂O₄ phase. The cubes of catalyst N5 have an edge size of 200–400 nm and are perhaps α -Al₂O₃ phase and thoroughly ripened ZnAl₂O₄ phase. The tiny particles of catalyst N5 have a size of only 10–20 nm and are probably iron species.

As mentioned above, the high temperature annealing at 1200 °C is conventionally utilized in the literature [13,14] to transform alumina phases to α -Al₂O₃ phase, by which the surface acidic centers may be greatly suppressed through the elimination of surface hydroxyl groups and, simultaneously, the surface basic centers may be adjusted to regulate the selectivities of products. Thus, this research employs high temperature annealing at 1200 °C to form solid-base type supports that carry iron/potassium particles, which may probably generate selective solid-base type FTO catalysts.

3.4. Surface basicity

M.E. Dry et al. point out that carbon monoxide will tend to pull electrons from iron when it adsorbs onto the surface of iron, while hydrogen will tend to force electrons towards iron as it adsorbs onto the surface of iron [31]. They also disclose that a solid base can push electrons to iron, strengthen Fe–C bond, and weaken the hydrogenation action [31]. Therefore, we focus on the investigation of solid-base type iron-based FTO catalysts. The tests of CO₂-TPD are conducted to determine the surface basicity of iron-based FTO catalysts.

The CO₂-TPD data of catalyst N5, shown in Fig. 5 & S8 and Table S2, give two CO₂ desorption peaks: one at 186.0 °C ascribed to the desorption of weakly adsorbed CO₂, while the other at 732.4 °C assigned to the desorption of strongly adsorbed CO₂. Catalyst N5 has a total CO₂-TPD peak area of 36,394 (A.U.), which is significantly larger than the

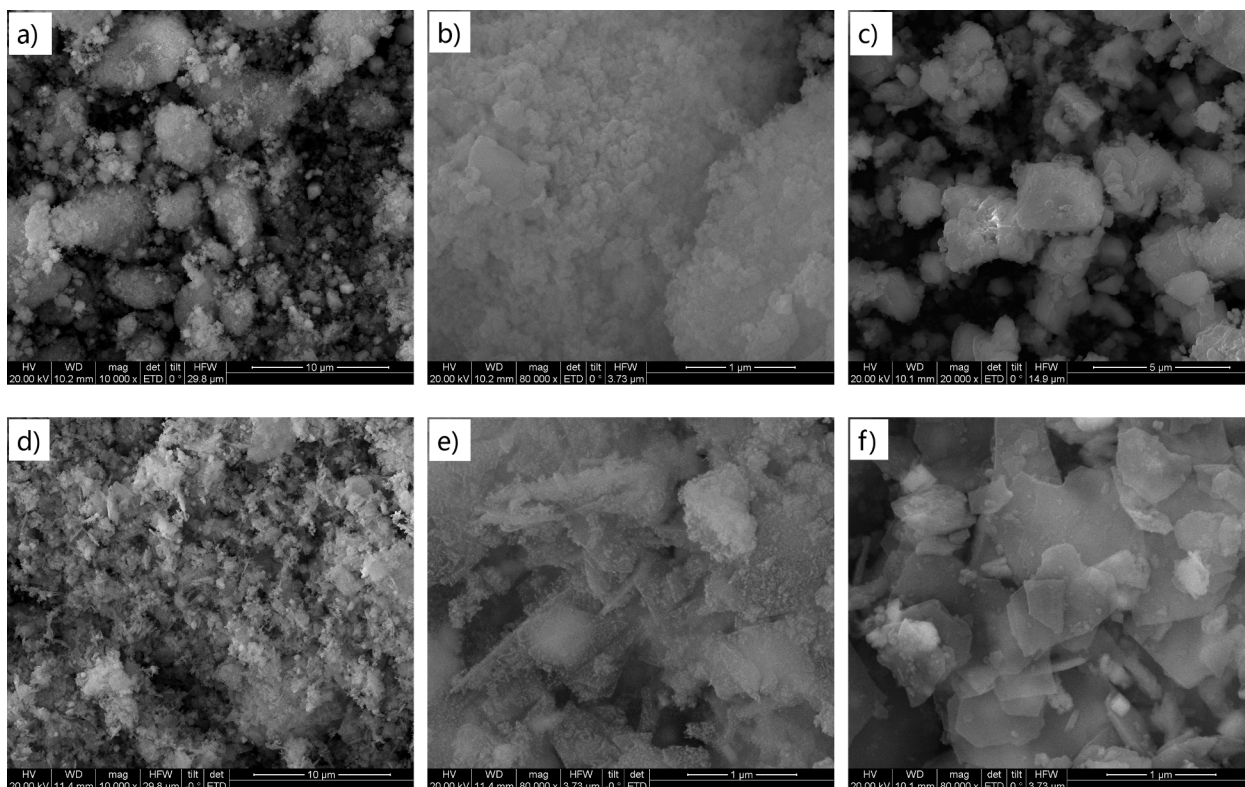


Fig. 4. FESEM images of freshly reduced catalysts. a) & b): catalyst N1. c): catalyst REF0. d) & e): catalyst N5. f): catalyst REF ∞ .

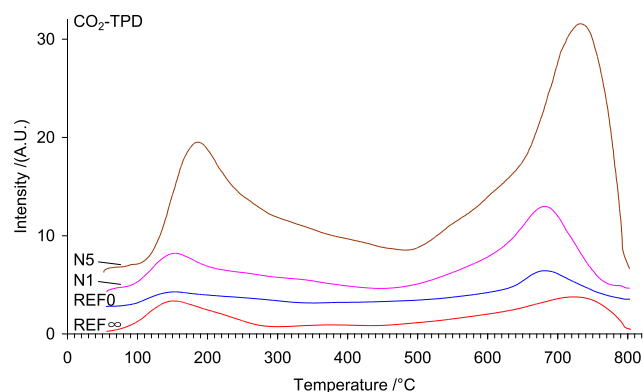


Fig. 5. CO₂ temperature-programmed desorption (CO₂-TPD) of freshly reduced catalysts.

peak areas of all other catalysts. The area of high temperature peak of catalyst N5 is significantly larger than that of its low temperature peak, whilst the intensity of high temperature peak of catalyst N5 is significantly more intense than that of its low temperature peak.

The CO₂-TPD data of catalyst N1, shown in Figs. 5, S6 & S8 and Table S2, display two CO₂ desorption peaks: one at 154.1 °C ascribed to the desorption of weakly adsorbed CO₂, while the other at 681.9 °C assigned to the desorption of strongly adsorbed CO₂. Catalyst N1 has a total CO₂-TPD peak area of 17,104 (A.U.). The area of high temperature peak of catalyst N1 is also significantly larger than that of the low temperature peak, whilst the intensity of high temperature peak of catalyst N1 is also significantly more intense than that of the low temperature peak.

The CO₂-TPD data of reference catalyst REF0, shown in Figs. 5, S6 & S8 and Table S2, provide two CO₂ desorption peaks: one at 152.8 °C ascribed to the desorption of weakly adsorbed CO₂, while the other at 680.5 °C assigned to the desorption of strongly adsorbed CO₂. Catalyst REF0 has a total CO₂-TPD peak area of 8936 (A.U.), which is significantly smaller than the peak areas of all other catalysts. The area of high temperature peak of catalyst REF0 is significantly larger than that of its low temperature peak, whilst the intensity of high temperature peak of catalyst REF0 is significantly more intense than that of its low temperature peak.

The CO₂-TPD data of reference catalyst REF∞, shown in Figs. 5, S6 & S8 and Table S2, offer two CO₂ desorption peaks: one at 152.0 °C ascribed to the desorption of weakly adsorbed CO₂, while the other at ca. 722.4 °C assigned to the desorption of strongly adsorbed CO₂. Catalyst REF∞ has a total CO₂-TPD peak area of 11,330 (A.U.). The area of high temperature peak of catalyst REF∞ is significantly larger than that of its low temperature peak, while the intensity of high temperature peak of catalyst REF∞ is only slightly more intense than that of its low temperature peak.

Theoretically, the K₂CO₃ added in the preparation of catalysts is possible to release CO₂ gas through thermal decomposition and thus may interfere with the detection of CO₂-TPD. However, in fact, the starting temperature of thermal decomposition for K₂CO₃ at atmospheric pressure is its melting point of 891 °C [32], so it will not interfere with these CO₂-TPD tests in the temperature range of 50 ~ 800 °C.

3.5. Catalytic performance tests

3.5.1. Catalytic activity

The CO conversions of catalysts are shown in Fig. 6 and Supporting Information Tables S3~S6, all exhibiting a positive correlation with their reaction temperatures. It is an often employed means to compare the catalytic activity using the deduced reaction temperatures as deduced CO conversions reach a value of 50% [26]. A catalyst may be regarded as a less active one if it needs a higher reaction temperature to

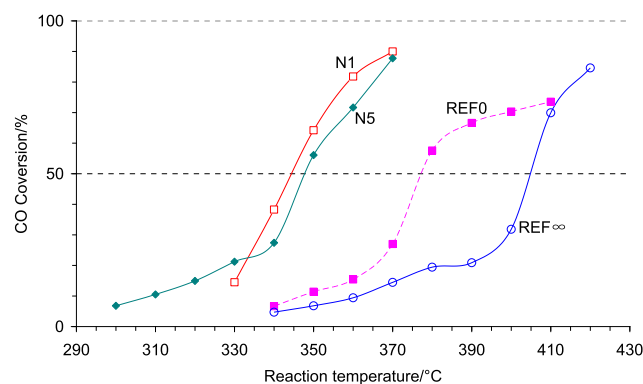


Fig. 6. Comparison of CO conversions for reduced catalysts at different reaction temperatures.

attain a CO conversion of 50% [26]. By utilizing this regulation for activity comparison, the catalytic activity of catalysts may be derived to have following degressive sequence of N1 > N5 > REF0 > REF∞. At a common reaction temperature of 360 °C, the FTY (iron time yield, 10⁻⁵ mol_{CO}/g_{Fe} s) values of catalysts show a decreasing sequence of N1(7.0) > N5(6.1) > REF0(1.3) > REF∞(0.8). It is clear that, with the contrast of CO conversion and FTY data of all four catalysts, linear relationship between catalytic activity and the $\times$ value is not observed. However, if the pure ZnAl₂O₄-supported catalyst (i.e., REF0) is not counted, then the catalytic activities of three alumina-containing catalysts (i.e., N1, N5, REF∞) exhibit a negative correlation with their $\times$ values.

3.5.2. Selectivity to CO₂

Theoretically, the selectivity to carbon dioxide for CO hydrogenation should be lower than 50%. Actually, other than the selectivity to CO₂ (51.7%) of catalyst N1 at a CO conversion of 64.2%, all selectivities to CO₂ of four catalysts obey this regulation and are below 50% (Tables S3~S6). Basing on an important opinion of the literature, this type of high CO₂ selectivity beyond 50% indicates that catalyst N1 may be functional in direct dissociation of carbon monoxide which releases oxygen atoms as carbon dioxide, or effective in water-gas shift reaction which converts some CO molecules to CO₂ molecules [33].

3.5.3. HC selectivity (C₅₊)

HC selectivity may be also denoted as hydrocarbon selectivity, hydrocarbon product distribution, or CH selectivity. HC selectivity is based on C molar content, shown by Fig. 7~11&S7 as well as Tables S3~S6, and comprises the selectivities of C₁, C₂-C₄, C₅₊ and so forth within hydrocarbon products. Each type of HC selectivity is expressed respectively with different reaction temperature (Fig. 7a~11a & S7a) and different CO conversion (Fig. 7b~11b & S7b).

For general Fischer-Tropsch reactions over iron or cobalt-based catalysts, higher reaction temperature will usually expedite the desorption rate of primary products and, therefore, decrease the C-C coupling opportunity and cut down the hydrocarbon selectivity of C₅₊ product. On the other side, lower reaction temperature will usually allow a relatively longer residence time for primary products and, therefore, increase the C-C coupling opportunity and enhance the hydrocarbon selectivity of C₅₊ product. As shown in Fig. 7, the case of alumina-supported Fe/K reference catalyst (i.e., REF∞) perfectly follows these rules, having a gradual decrease of C₅₊ HC selectivity from 70.9% to 13.4% as the temperature increases from 340 to 420 °C. The ZnAl₂O₄-supported Fe/K reference catalyst (i.e., REF0) only follows these rules in the temperature range of 340 ~ 370 °C with a gradual decrease of C₅₊ HC selectivity from 67.0% to 53.9%. At higher temperatures above 370 °C, the C₅₊ HC selectivity of catalyst REF0 fluctuates in a narrow range of 53.5% ~ 57.9%. This fact indicates that, compared with alumina-supported Fe/K catalyst (i.e., REF∞), this

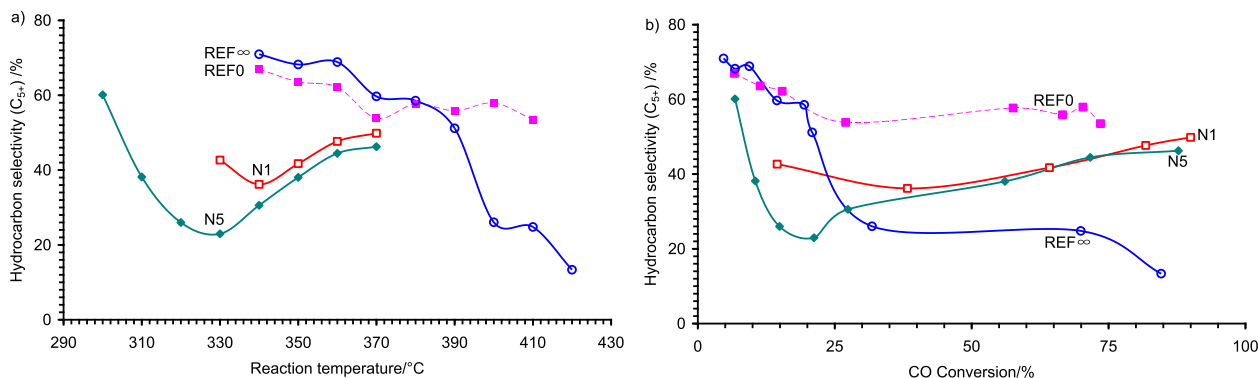


Fig. 7. C_{5+} hydrocarbon distributions (i.e., C_{5+} HC selectivity) of reduced catalysts at different reaction temperatures (a) and different CO conversions (b).

ZnAl₂O₄-supported Fe/K catalyst (i.e., REF0) is able to attach the primary products to stay on its surface for a longer time at higher temperatures above 370 °C, make the primary products to undergo more carbon chain propagation reactions, and thus be able to resist against the regular decrease of C_{5+} HC selectivity at elevated temperature range beyond 370 °C. These data reveal the significant difference in function between ZnAl₂O₄ support and alumina support.

The C_{5+} HC selectivity curves of catalysts N5 and N1 look bizarre but are similar to each other, both having a degressive trend first, and then, after reaching their individual minimum values, both exhibiting an ascensive trend. At the CO conversion range below 21.2%, the decrease of catalyst N5's C_{5+} HC selectivity curve is more abrupt than both catalysts REF ∞ and REF0, indicating that, besides the temperature factor, the coexistence of zinc aluminate and alumina may generate a synergistic effect that makes catalyst N5 effectively repel primary products away from the surface and thus reach a sharp drop of C_{5+} HC selectivity. As the CO conversion exceeds 21.2% where the reaction temperature is beyond 330 °C, above mentioned function of ZnAl₂O₄ phase starts to boost, holding primary products for chain propagation reactions over catalyst surface and, consequently, gearing up the C_{5+} HC selectivity even at elevated temperatures. For the case of catalyst N1, this turning point comes at slightly higher temperature of 340 °C, together with a CO conversion of 38.3%.

3.5.4. HC selectivity (C_1 - C_4)

The C_1 - C_4 HC selectivity is the value of 100% minus C_{5+} HC selectivity. At lower CO conversions below 25%, the C_1 - C_4 HC selectivity values of catalysts N5 and N1 are higher than those of catalysts REF ∞ and REF0 (Fig. S7). The C_1 - C_4 HC selectivity of catalyst REF ∞ becomes higher than others at 420 °C, which is not welcome because a catalyst is prone to deactivate at high CO conversion due to the sintering and formation of inactive metal-support compounds in high temperature steam environment. If the C_1 - C_4 HC selectivity becomes low, as a portion

of C_1 - C_4 HC selectivity, then the C_2 - C_4 HC selectivity consequently will not be high.

3.5.5. HC selectivity (C_2 - C_4)

All C_2 - C_4 HC selectivity curves (Fig. 8) of catalysts N1, N5, REF0 and REF ∞ exhibit resembling tendencies to their corresponding C_1 - C_4 HC selectivity curves (Fig. S7), presenting that the content of C_2 - C_4 products is positively related with the catalyst's capability of weakening chain propagation. Once the C-C coupling is suppressed, the C_{5+} HC selectivity ought to be low and, consequently, the short chain product (i.e., C_1 and C_2 - C_4) selectivity shall become high, and vice versa. The C_2 - C_4 HC selectivity of catalyst N1 reaches a maximum value of 56.1% at 340 °C, while that of catalyst N5 attains a maximum value of 52.2% at 330 °C.

3.5.6. HC selectivity (CH_4)

The shapes of CH_4 HC selectivity curves (Fig. 9) of catalysts N5 and REF ∞ are still similar to the shapes of their corresponding C_1 - C_4 HC selectivity curves (Fig. S7), while those of catalysts N1 and REF0 are not. The C_1 - C_4 HC selectivity curve of catalyst N1 looks only a little lower than that of catalyst N5 (Fig. S7), while the CH_4 HC selectivity curve of catalyst N1 is significantly lower than that of catalyst N5 (Fig. 9). This fact presents that catalyst N1 ought to be more efficient in suppressing hydrogenation than catalyst N5, because CH_4 is not only a portion of C_1 - C_4 hydrocarbons but also a thoroughly hydrogenated product that needs catalytic hydrogenation ability. Methane is an unfavorable Fischer-Tropsch product. Even at its highest CO conversion, the CH_4 HC selectivity of catalyst N1 still remains below a surprisingly low value of 9%.

3.5.7. C_2 - C_4 olefin/paraffin ratio

During CO hydrogenation over iron or cobalt-based catalysts, the primary products are basically linear 1-*n*-alkenes. Then these primary products may undergo secondary reactions such as hydrogenation or/and further C-C coupling, generating some linear alkanes/alkenes

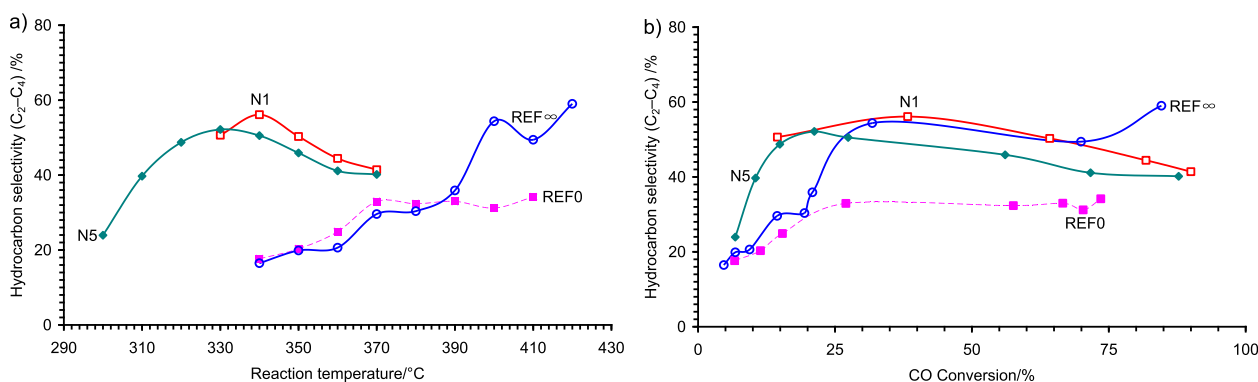


Fig. 8. C_2 - C_4 hydrocarbon distributions (i.e., C_2 - C_4 HC selectivity) of reduced catalysts at different reaction temperatures (a) and different CO conversions (b).

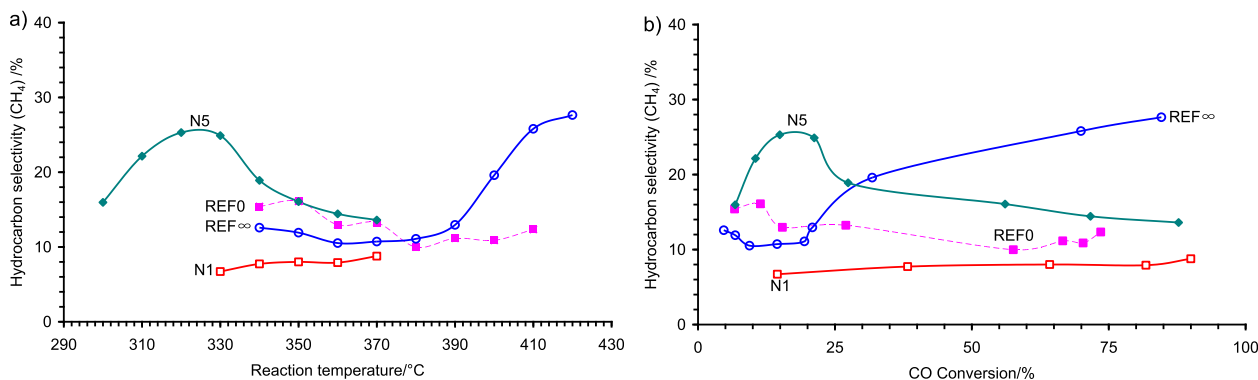


Fig. 9. CH₄ hydrocarbon distributions (i.e., CH₄ HC selectivity) of reduced catalysts at different reaction temperatures (a) and different CO conversions (b).

(probably with longer carbon chains) [34]. The C₂-C₄ olefin/paraffin ratio is thus positively correlated with catalyst's ability of suppressing hydrogenation.

Lower CO conversion is more practical than higher CO conversion due to more serious decay at high CO conversions, at which the high H₂O partial pressure may cause the formation of inactive metal-support compounds in harsh steam environments and thus expedite the process of catalyst deactivation [35]. The C₂-C₄ olefin/paraffin (O/P) ratios of catalysts exhibit a decreasing order of N1 > REF0 ≈ REF ∞ > N5 at lower CO conversions below 32% (Fig. 10b and Tables S3 ~ S6). These data indicate that catalyst N1 owns significantly more powerful ability of suppressing hydrogenation than other three catalysts.

Fischer-Tropsch reaction is exothermic and the elevation of reaction temperature will accelerate the desorption rate of intermediate species, which will suppress the secondary reactions such as chain propagation and secondary hydrogenation. Therefore, at higher temperatures, the hydrogenation ability of a catalyst ought to be less powerful and the C₂-C₄ O/P ratio ought to be higher at higher temperatures and consequently higher CO conversions. However, at higher CO conversions beyond ca. 50%, catalysts N1, REF0 and REF ∞ have a decreasing trend in their C₂-C₄ O/P ratios (Fig. 10b). The C₂-C₄ O/P ratio of catalyst N5 does not show obvious decrease even at a high CO conversion of 87.8%. These facts indicate that, with the highest amount (i.e., represented by A_{total}) and the intense strength (i.e., represented by CO₂ desorption temperature) of basicity (Table S2), the ability of suppressing hydrogenation of catalyst N1 becomes steady and almost unchangeable at higher temperatures and higher CO conversions.

The strength of basic sites may be expressed by the desorption temperatures of CO₂-desorption peaks. A higher CO₂ desorption temperature means a stronger basicity strength. As shown in Figs. 5 & 10b and Table S2, the C₂-C₄ O/P ratio is influenced by the surface basicity data. The catalysts with stronger strength of basic sites may have lower C₂-C₄ O/P ratios than the catalysts with weaker strength of basic sites, as

displayed by the catalysts (i.e., N1 vs. N5) with a dual-phase support in the whole CO conversion range or by the catalysts (i.e., REF0 vs. REF ∞) with a mono-phase support in the CO conversion range above 31.8%. Correlations between the C₂-C₄ O/P ratios of all four catalysts and their values of $I_{\text{high}}/I_{\text{low}}$ or $A_{\text{high}}/A_{\text{low}}$ (Table S2) remain unclear. The value of $I_{\text{high}}/I_{\text{low}}$ denotes the ratio of integral intensity of high temperature peak to that of low temperature peak in a CO₂-TPD plot, while the value of $A_{\text{high}}/A_{\text{low}}$ denotes the ratio of integral area of high temperature peak to that of low temperature peak in a CO₂-TPD plot.

Moreover, at higher CO conversions above ca. 50%, reference catalyst REF ∞ shows the lowest C₂-C₄ O/P ratio than other three catalysts. These data correspond to the carburization degree of iron in catalysts (Table 1). The condition of iron in used catalyst REF ∞ is X-ray amorphous while the conditions of iron in other three catalysts are all crystalline Fe₅C₂ phase, indicating that the C₂-C₄ O/P ratio may be related to the carburization degree of iron. At the whole CO conversion range, catalyst N1 owns very high C₂-C₄ O/P ratio of 9.2–13.0, which may provide an opportunity for catalyst N1 to generate a considerable C₂-C₄ HC selectivity together with a very low methane HC selectivity.

3.5.8. Hydrocarbon selectivity (C₂-C₄)

The C₂-C₄ HC selectivity equals to the product of a multiplier (i.e., the C₂-C₄ HC selectivity) and a multiplicand (i.e., the C₂-C₄ O/P ratio). Therefore, as shown in Figs. 11 and 8, the C₂-C₄ HC selectivities of all catalysts look a little similar to their C₂-C₄ HC selectivities.

At a total pressure of 20 bar and a temperature of 340 °C, the C₂-C₄ HC selectivity of famous S/Na promoted Fe-based FTO catalyst reaches a maximum value of 51% with a C₂-C₄ O/P ratio of 2.7 and a methane HC selectivity of 19% [14]. At a total pressure of 2.0 MPa (≈ 20 bar) and a temperature of 340 °C, the C₂-C₄ HC selectivity of catalyst N1 achieves its maximum value of 52.1% with a very high C₂-C₄ O/P ratio of 13.0 and a very low methane HC selectivity of only 7.7%. At a C₂-C₄ CH selectivity of 52.1% and a CO conversion of 38.3%, catalyst N1 presents

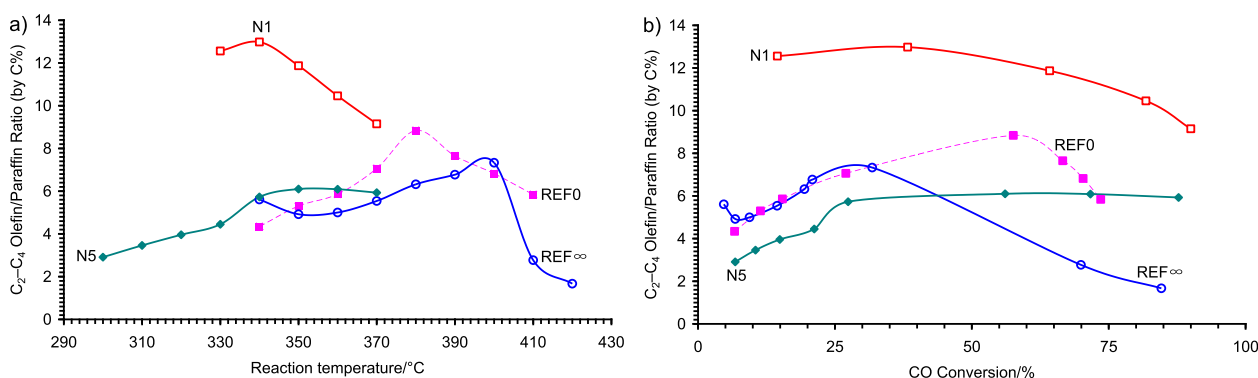


Fig. 10. C₂-C₄ olefin/paraffin ratios of reduced catalysts at different reaction temperatures (a) and different CO conversions (b).

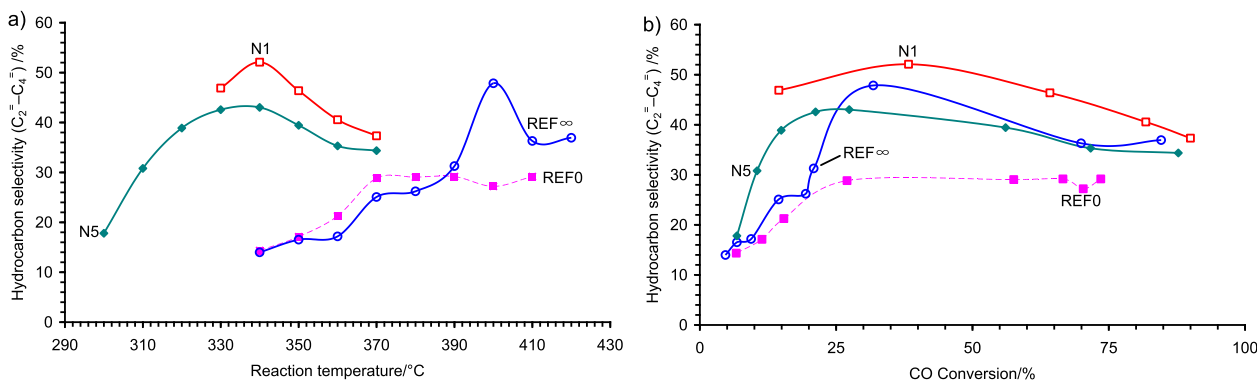


Fig. 11. C_2-C_4 hydrocarbon distributions (i.e., C_2-C_4 CH selectivity) of reduced catalysts at different reaction temperatures (a) and different CO conversions (b).

its C_2 , C_3 , and C_4 O/P ratios of 9.8, 17.7, and 13.0, respectively. With the advantage of such high O/P ratios, the contents of ethylene in C_2 (also denoted as C_2 olefin fraction), propylene in C_3 , and butylene in C_4 hydrocarbons of catalyst N1 respectively accomplish super high values of 90.7%, 94.7%, and 92.9% at 340 °C and ca. 20 bar. A lifetime test, i.e., the evolution of the conversion and selectivities (including CO_2 selectivity) vs. time on stream, will be conducted in our future work.

4. Conclusions

A series of catalysts are synthesized by loading potassium and iron over high temperature calcined supports with nominal compositions of $ZnAl_2O_4 \cdot xAl_2O_3$ ($x = 1, 5$) and are examined together with reference catalysts ($x = 0, \infty$). The catalytic activities of three alumina-containing catalysts exhibit a negative correlation with their x values.

Alumina and $ZnAl_2O_4$ phases in catalyst supports create different effects upon the C_{5+} hydrocarbon selectivities of catalysts. Generally, a catalyst with relatively weaker strength of basic sites may own a relatively higher C_2-C_4 olefin/paraffin ratio. The C_2-C_4 O/P ratio is also influenced by the carburization degree of iron, where Fe_5C_2 may generate higher C_2-C_4 O/P ratio than X-ray amorphous iron species.

The hydrogenation ability of catalyst N1 has been successfully suppressed. At a total pressure of 2.0 MPa (≈ 20 bar), the C_2-C_4 HC selectivity of catalyst N1, 15%Fe/2%K₂O/83% ($ZnAl_2O_4 \cdot 1Al_2O_3$), reaches its maximum value of 52.1% at 340 °C and a CO conversion of 38.3%, whilst its contents of ethylene in C_2 , propylene in C_3 , and butylene in C_4 hydrocarbons achieve ultra-high values of 90.7%, 94.7%, and 92.9%, respectively. Meanwhile, this catalyst exhibits a very significant C_2-C_4 O/P ratio of 13.0 and a very low methane HC selectivity of only 7.7%.

CRediT authorship contribution statement

Zhenxin Liu: Supervision, Methodology, Data curation, Formal analysis, Validation, Writing - original draft. **Gaopeng Jia:** Investigation. **Chenxi Zhao:** Investigation. **Yu Xing:** Conceptualization, Funding acquisition, Project administration, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fuel.2020.119572>.

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